when is it your turn?

digitalisierung und programmierung · prof. dr. nicolas meseth

1. why it collapses
2. the clocked way
3. the clock-free way

part 1

why it collapses

how do eighty people stay together?

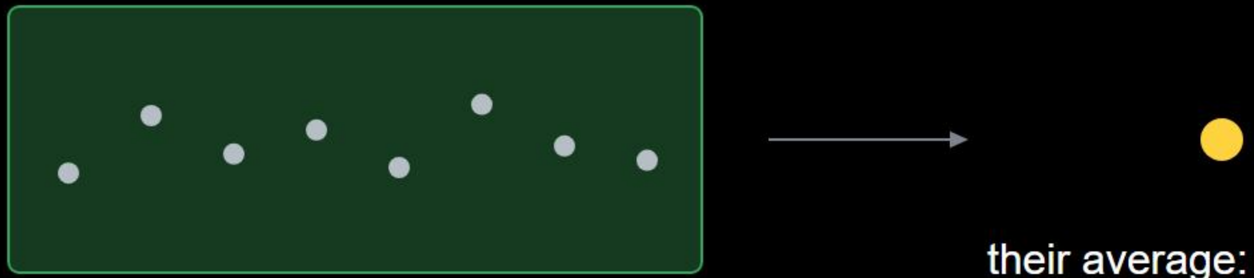
image: ai-generated (gpt-image-2)

your homework: where did it tip?

team	symbols per second	what broke
team 7	14	letters shifted by one, slowly, after about a minute
team 2	9	suddenly nothing matched a profile any more
team 11	6	every second word was garbled from the start
team 5	4	held, we did not dare go faster

same limit for everyone. very different numbers.

one symbol, zoomed in

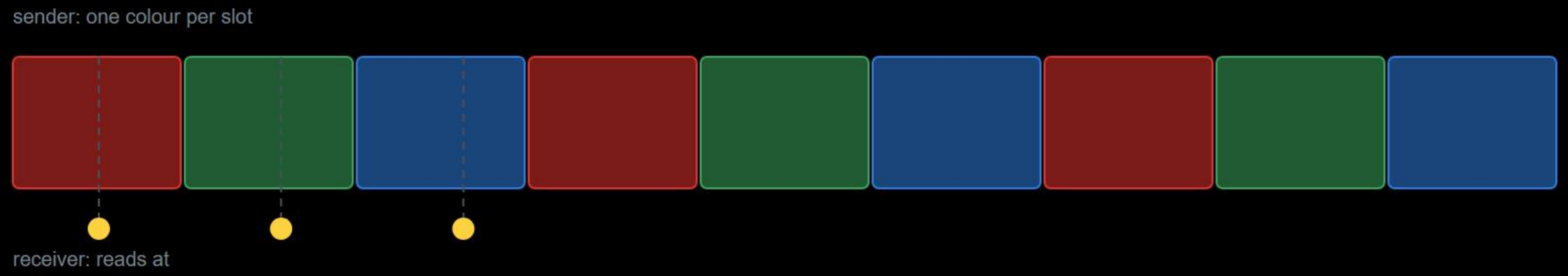


eight quick readings, each a little different

their average: one value,
the dot in every picture from here on

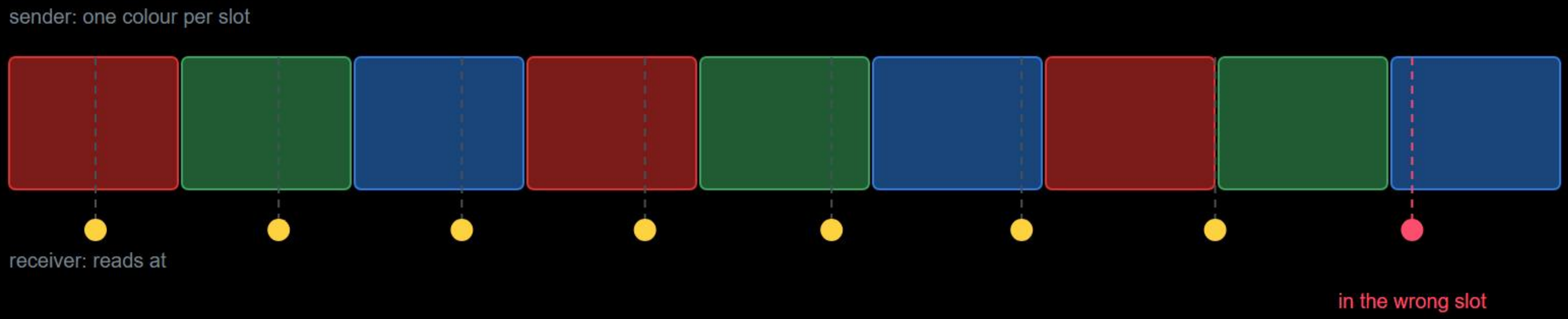
one dot in the pictures = many readings, averaged.

cause one: no two clocks run alike



the tiny error per symbol is not the problem. the sum is.

cause one: no two clocks run alike



the tiny error per symbol is not the problem. the sum is.

cause two: the sensor needs time to look

symbols 100 ms long

symbols 40 ms long

the reading falls inside one colour

matches one of your profiles

every reading holds a colour change

matches nothing

no error, no warning, just a meaningless number.

both causes are one question

when does a new symbol begin?

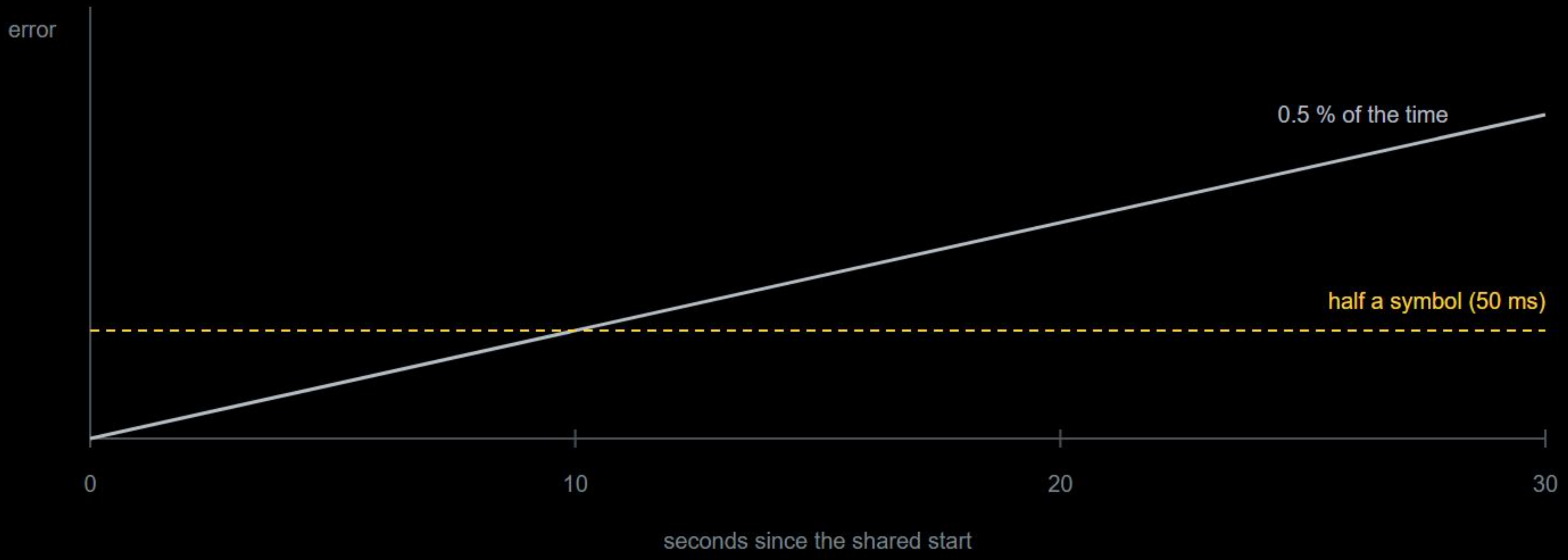
answer one: time tells you.

answer two: change tells you.

part 2

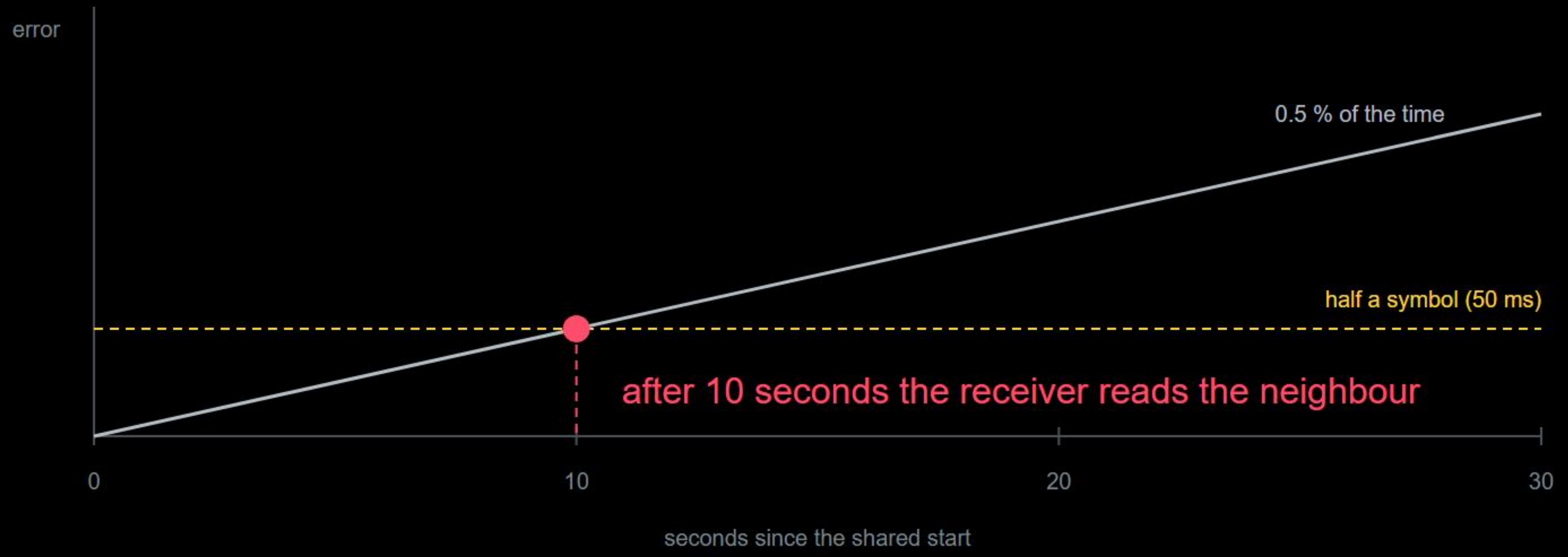
the clocked way

when does it tip?



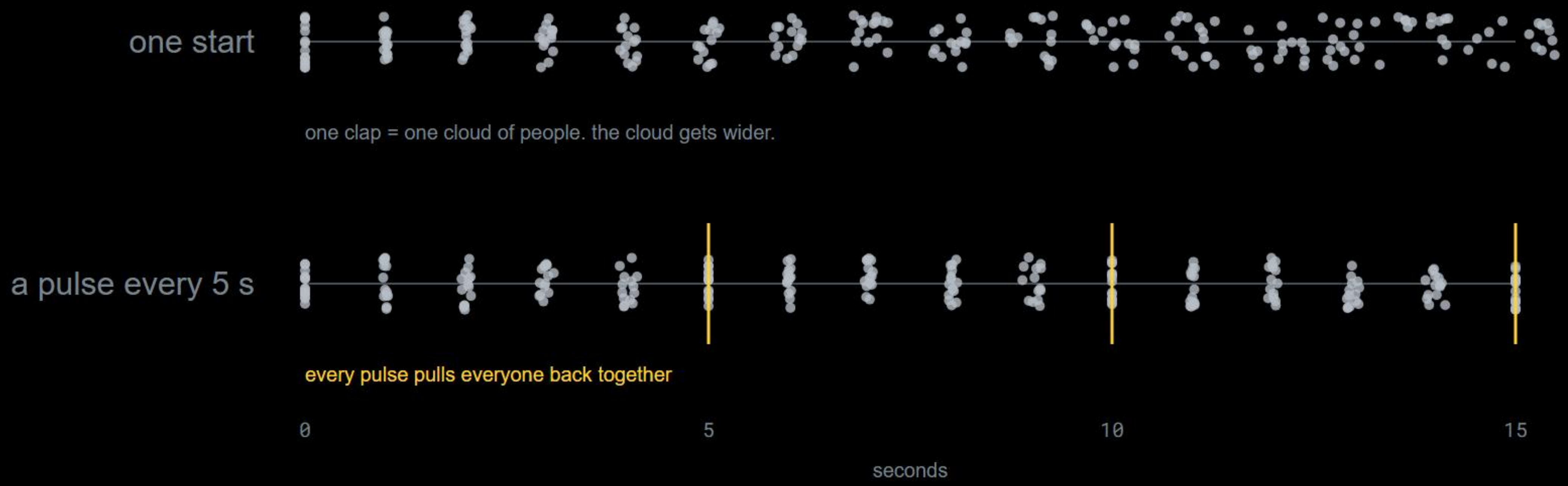
a shared start is not a question of if it tips, only when.

when does it tip?



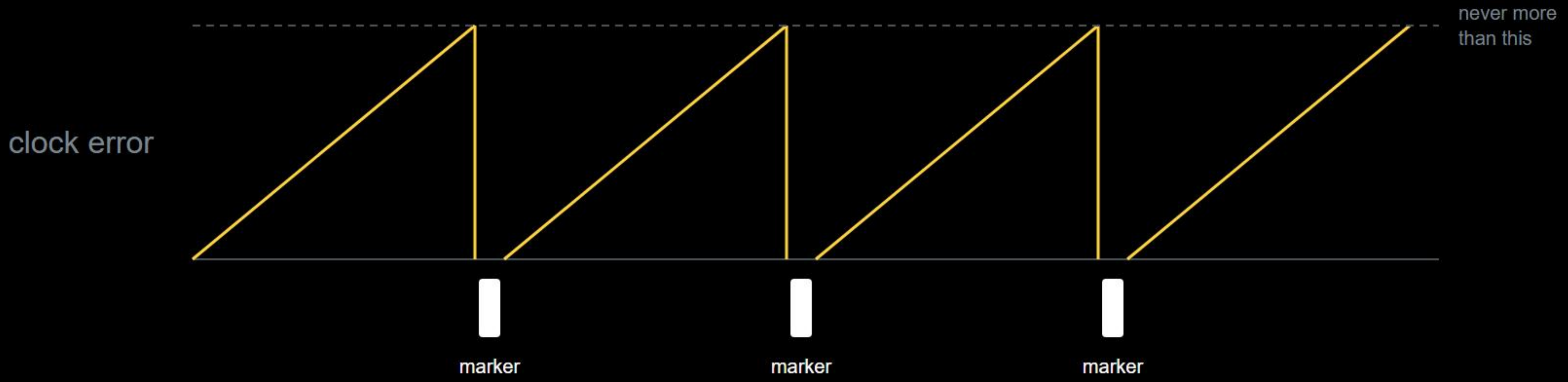
a shared start is not a question of if it tips, only when.

eighty people, one clap



agreeing once is not enough. agreeing again and again is.

recurring markers reset the drift



a marker is a symbol reserved for this job: a colour outside your data alphabet, or a short dark pause

the error can never grow beyond one block.

try it: the drift simulator

:sent

:received

simulated clocks: real crystals differ by fractions of a percent. the effect is the same, only slower.

receiver clock	currently off by	half a slot off after	wrong reads	marker overhead
+2 %	+0.00 slots	25 sym	0	0 %
runs slow: samples too late	half a slot means wrong reads	then reads start to miss	of the last 100 symbols	slots spent on markers

receiver clock error

2 % per slot

sync marker every … symbols

off: no resync

display speed

40 % of real time

pause | sample once | send marker | reset the price of the clocked way

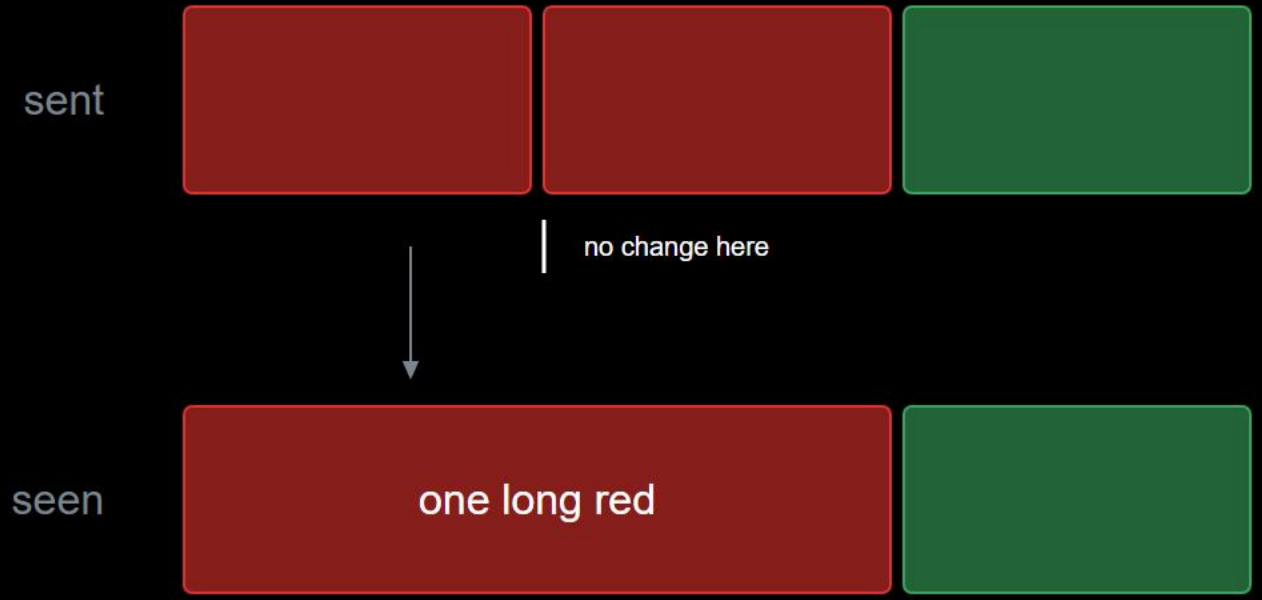
start once	recurring markers
not a single symbol spent	drift stays bounded
drift adds up without limit	markers carry no message

how bad are your clocks? that is a measurement, not a belief.

part 3

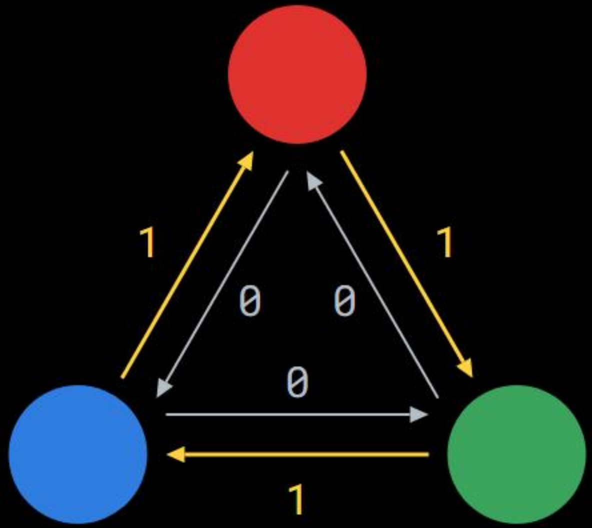
the clock-free way

first idea: a change means "next symbol"



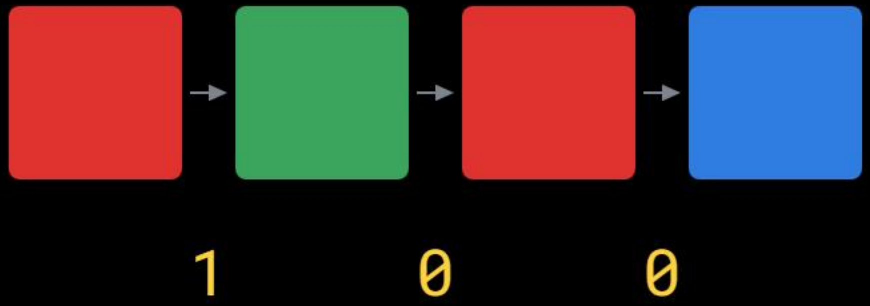
no change, no symbol. "red red" arrives as one.

the fix: the transition carries the bit



clockwise: 1, anticlockwise: 0

your own colour is never allowed next



measure as often as you like. decode only on change.

freedom is paid in capacity

bits carried by one symbol

k	as states	as transitions
3	1.58	1.00
4	2.00	1.58
5	2.32	2.00

five colours,
four allowed successors,
a clean 2 bits

want a clean 2 bits per transition? you need five colours, not four.

the most common wrong conclusion

clock-free is not time-free

how long must a colour hold to count?

how many transitions per second can the sensor follow?

$$t = n / (r * b)$$

n	bits to send
r	symbols per second
b	bits per symbol

two levers, nothing else: bits per symbol, symbols per second.

$$t = n / (r * b)$$

n	bits to send
r	symbols per second
b	bits per symbol

16 000 bits, 4 colours, 5 symbols per second

clocked	`b = log2(k) = 2.00`	about 27 minutes
clock-free	`b = log2(k-1) = 1.58`	about 34 minutes

two levers, nothing else: bits per symbol, symbols per second.

today

1 the clapping experiment

2 workshop: find your maximum rate, with evidence

3 write down which way you lean, clocked or clock-free, and why

no two clocks agree.
the only choice is what you do about it.

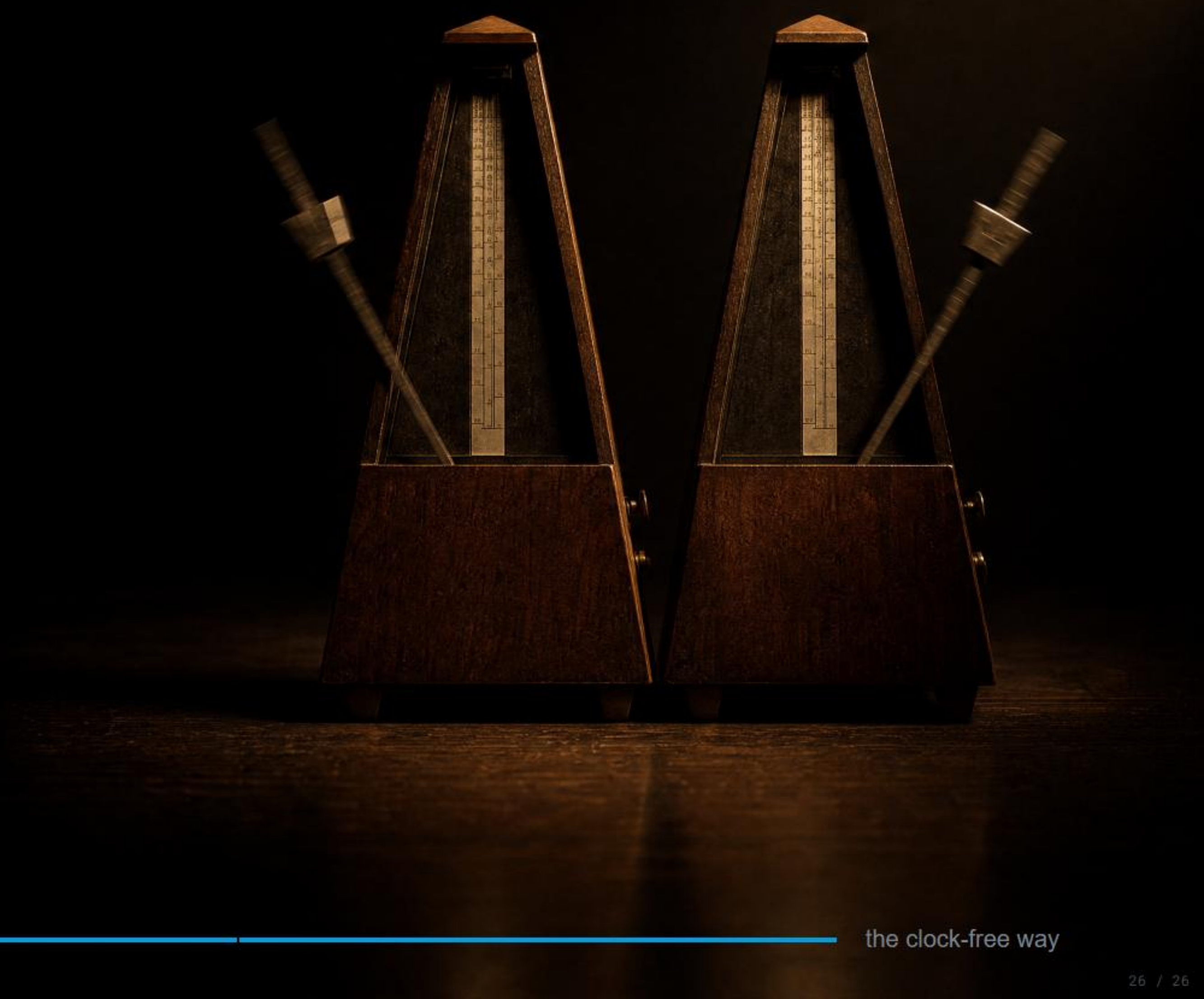


image: ai-generated (gpt-image-2)